



FEB 18 – USING LLMSTHROUGH API

WHY WOULD I USE LLMS VIA CHATBOX INTERFACE INSTEAD OF API?

What does API mean?

Application Programming Interface

CLAUDE: REASONS TO USE LLMS VIA API VS. CHAT INTERFACE

Automation & Integration

- Workflow integration — Embed LLM calls into scripts, apps, pipelines, and existing tools
- **Batch processing** — Process hundreds or thousands of prompts programmatically
- **Chaining tasks** — Pipe outputs into other systems or subsequent LLM calls automatically

Control & Customization

- System prompts — Set precise, persistent instructions for model behavior
- **Parameter tuning** — Adjust temperature, top-p, max tokens, stop sequences, etc.
- **Model selection** — Choose specific model versions for consistency and reproducibility
- Structured output — Request JSON or other formats more reliably

Cost & Efficiency

- **Pay-per-token pricing** — Can be cheaper for high-volume or short-query use cases
- **No subscription commitment** — Pay only for what you use
- Speed — No UI overhead; responses can stream directly into your code

Building Products

- Power user-facing applications — Chatbots, copilots, content generators, etc.
- Multi-user access — Serve many users under one account
- Custom UIs — Build interfaces tailored to your specific needs

Data & Privacy

- **Greater control over data handling** — Depending on the provider's API terms
- Keep inputs/outputs in your own infrastructure
- Logging and auditing on your terms

Limitations to Be Aware Of

- Requires **programming knowledge**
- You manage error handling, rate limits, and retries
- **No built-in conversation memory (you manage context yourself)**
- **API pricing can exceed subscription cost** for heavy casual use

In short: The chat interface is great for **interactive, ad hoc** use. The API shines when you need **scale, automation, customization, or integration** into larger systems.

GEMINI – VERY SIMILAR OUTPUT

Feature	Chatbox (Web UI)	API Key
Primary Use	Human-to-AI Conversation	App-to-AI Automation
Pricing	Flat Monthly Fee (usually)	Pay-per-token (Usage based)
Privacy	Often trains on your data	usually Zero Data Retention
Output	Conversational Text	Text, JSON, Function Calls
Speed	One message at a time	Parallel processing
Memory	Managed by the AI provider	Managed by you (Context Window)

WHERE DO WE CHECK COST?

Model pricing

The following table shows pricing for all Claude models across different usage tiers:

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.6	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.5	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.1	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Opus 4	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Sonnet 4.5	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 3.7 (deprecated)	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Haiku 4.5	\$1 / MTok	\$1.25 / MTok	\$2 / MTok	\$0.10 / MTok	\$5 / MTok
Claude Haiku 3.5	\$0.80 / MTok	\$1 / MTok	\$1.6 / MTok	\$0.08 / MTok	\$4 / MTok
Claude Opus 3 (deprecated)	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Haiku 3	\$0.25 / MTok	\$0.30 / MTok	\$0.50 / MTok	\$0.03 / MTok	\$1.25 / MTok

input much cheaper
than output (about 5x)

Fast mode pricing

Fast mode for Claude Opus 4.6 (research preview) provides significantly faster output at premium pricing (6x standard rates). Currently supported on Opus 4.6:

Context window	Input	Output
≤ 200K input tokens	\$30 / MTok	\$150 / MTok
> 200K input tokens	\$60 / MTok	\$225 / MTok

be wary of special features (e.g.
“fast” inference)

COMPARISON TABLE (LEGEND FOR MODEL NAMES)

Provider	Category	Flagship model name	Everyday model name	“Research/Deep” model
OpenAI	Text generation	gpt-5.2 (flagship)	gpt-5-mini (everyday)	o3-deep-research
	Image (token eq.)	gpt-image-1.5	gpt-image-1-mini	–
Anthropic	Text generation	claude-opus-4.5 (flagship)	claude-sonnet-4.5 (everyday)	(same tiers; no distinct extra token research tier)
	(cheapest)	claude-haiku-4.5 (ultra-low cost)		
Google Gemini	Text generation	gemini-3-pro (flagship)	gemini-2.5-flash (everyday)	gemini-2.5-flash-lite (ultra-cheap)
	Image/token eq.	gemini-2.5-flash-image	(same)	–
Perplexity	Text generation	sonar-pro (flagship)	sonar-base (everyday)	sonar-deep-research
Kimi (Moonshot)	Text generation	K2.5 (flagship)	–	K2-Thinking
	Image (token eq.)	moonshot-v1	–	–

COMPARISON TABLE – PRICES FOR INPUT/OUTPUT

Provider	Category	Flagship (\$/IM)	Everyday (\$/IM)	Research/Deep (\$/IM)
OpenAI	Text	\$1.75 / \$14.00	\$0.25 / \$2.00	\$10 / \$40
	Image	~\$8 / ~\$32	~\$2.5 / ~\$8	—
Anthropic	Text	\$5 / \$25	\$3 / \$15	(same tier)
	Low-cost	\$1 / \$5	—	—
Google Gemini	Text	~\$1.25 / ~\$10	~\$0.30 / ~\$2.50	~\$0.10 / ~\$0.40
	Image	~\$0.30 (input) + output eq ~\$30 per IM	(same)	—
Perplexity	Text	\$3 / \$15	\$0.25 / \$2.50	\$2 / \$8 (+ extras)
Kimi (Moonshot)	Text	\$0.60 / \$3.00	—	\$0.60 / \$2.50
	Image	~\$0.20 / \$2.00	—	—

HOW MANY TOKENS ARE 1,000 TOKENS?

- In my LLM pricing conversation, I had 4-5 short prompts and a bunch of outputs, so most tokens were output tokens.
- I was using GPT 5.2 (**flagship; I didn't really have a choice**) and its estimate was that I used ~7,000–9,000 tokens. So, this simple conversation would have cost $\sim 8k * \$14 / 1M = \0.112 in API mode, although I could have used a less expensive model in that case: gpt5-mini would have been 7x cheaper and probably equally performant for this use case.
- Also, GPT erroneously quoted price per 1M tokens as price for 1k 😅

LET'S GET SOME API KEYS SET UP! (6 CLAUDE, 6 OPEN AI)

HOMEWORK (FOR FEB 25)

Re-do your task from last Wednesday using the API (in workbench or programming mode). Please for the love of all that's holy **ONLY USE THE MORE MODEST MODEL** (Claude haiku, gpt-5 mini) because despite placing all the boundaries, the APIs for these models don't stop immediately when we run out of money, they allow us (me) to go negative!

We are not doing this for performance, just to familiarize ourselves with the format and think about use cases where one may be preferable to the other (we'll see more of this in upcoming weeks, with coding). If your original task was too complex, don't worry, do step 1 or pick a simpler one.

Report briefly about your experience (pros and cons of one and the other method) in our shared doc (Experimenting with LLMs with APIs)